

# Breakdown of Core Departments in Major Tech Companies

The most core departments in major tech companies are: Ads Monetization, Recommendation/Search Algorithms, Core Product, Cloud and Infrastructure, AI/ML Platform, Data and Experimentation Platform, Growth, Commercialization Platform, Safety and Compliance, and Ecosystem.

Different types of companies place different levels of importance on different departments.

| Company Type           | Most Core Departments                                                  |
|------------------------|------------------------------------------------------------------------|
| Google / Meta / TikTok | Ads, Ranking, Feed/Search, Growth, AI Infrastructure                   |
| Amazon                 | AWS, Marketplace, Ads, Logistics, Prime, Recommendation                |
| Microsoft              | Azure, Microsoft 365, AI/Copilot, Developer Platform, Enterprise Sales |
| Apple                  | iPhone, iOS, Services, Silicon, App Store, Ecosystem                   |
| Netflix / YouTube      | Recommendation, Content, Membership, Ads, Streaming Infrastructure     |
| Nvidia                 | Data Center, AI Chips, CUDA Ecosystem, Enterprise AI Platform          |

## 1. Ads / Monetization Is the “Cash Cow Department” for Many Major Tech Companies

These departments are the closest to revenue, especially at Google, Meta, Amazon, and TikTok.

Google’s core revenue still comes from Google Services, especially Search, YouTube, advertising, and subscriptions. Alphabet’s 2025 annual report shows that Google Services revenue growth was mainly driven by Search, subscriptions/platforms/devices, and YouTube ads; YouTube ads and subscriptions generated more than \$60 billion in annual revenue.

Meta is an even more typical example. Meta clearly states in its annual report that almost all of its revenue comes from advertising within its Family of Apps, including Facebook, Instagram, Messenger, WhatsApp, and others.

Amazon is no longer just an e-commerce company. AWS is one of its core profit engines, and its advertising business has also become very large. In Amazon’s Q1 2026 disclosure, AWS operating income reached \$14.2 billion, and advertising services revenue continued to grow rapidly.

Therefore, inside major tech companies, departments such as Ads Engineering, Ads Ranking, Ads Measurement, Ads Marketplace, Ads Infrastructure, Conversion Modeling, Attribution, Bidding, Auction, and Privacy Ads are all extremely core areas.

## **2. Recommendation / Ranking / Search Algorithms Are the Engine of User Time and Revenue**

These departments are often more core than ordinary product teams because they directly determine:

What users see, what they click, what they buy, how long they stay, and how much advertising can be sold.

Typical departments include:

Search Ranking, Feed Ranking, Recommendation Systems, Candidate Generation, Retrieval, Personalization, Ads Ranking, Marketplace Ranking, Video Recommendation, and Notification Ranking.

For example, the core of TikTok/Douyin is its recommendation system. Public research has also mentioned that ByteDance's recommendation models have already been deployed in massive industrial-scale scenarios such as Douyin Ads and Douyin E-commerce.

Netflix, YouTube, TikTok, Instagram Reels, and Amazon product recommendations are all fundamentally driven by recommendation/ranking systems, which power both user experience and commercial revenue.

So your understanding of "recommendation and algorithm" is correct. This is one of the most technically difficult and strategically important core departments inside major tech companies.

## **3. Growth Is Core, but Growth Is Not Simply Marketing**

Growth inside major tech companies is not a traditional "marketing department." It is a department focused on data, experimentation, product, algorithms, and user lifecycle management.

Real Growth work usually includes:

User acquisition, activation, retention, reactivation, conversion, subscriptions, payments, referral growth, notification strategy, onboarding, pricing, membership, and funnel optimization.

For example, Meta, TikTok, Netflix, Amazon Prime, YouTube Premium, Microsoft 365, and Apple Services all have many Growth, Lifecycle, Subscription, and Engagement-related teams.

So your understanding of "growth/products" is correct, but one thing to note is:

Growth in major tech companies is usually not traditional marketing. It is Product Growth + Data Science + Experimentation + Machine Learning.

## **4. Product Is Core, but Not Every Product Team Is Core**

Inside major tech companies, not all product teams have the same level of importance.

The most core product teams are those that control the main user entry points, main revenue sources, and main user relationships. For example:

Google: Search, YouTube, Android, Chrome, Maps, Cloud  
Meta: Instagram, Facebook Feed, Reels, WhatsApp, Ads  
Amazon: Marketplace, Prime, Search, Checkout, AWS Console  
Microsoft: Azure, Microsoft 365, Teams, Windows, Copilot  
Apple: iPhone, iOS, App Store, Services, Wearables  
Netflix: Content Discovery, Playback, Membership, Ads

A peripheral product team may still be called a “Product” team, but that does not necessarily mean it is core. Truly core product lines usually have:

High DAU, high revenue, high strategic priority, CEO-level attention, and the ability to influence company valuation.

## **5. Cloud / Infrastructure Is One of the Most Core Foundational Departments Today**

This is something many people overlook.

In the AI era, Cloud, Compute, Storage, Networking, Distributed Systems, GPU Infrastructure, Data Centers, Serving Platforms, and ML Platforms have become some of the most core departments inside major tech companies.

Microsoft Azure has become one of Microsoft’s most important growth engines. Microsoft’s 2025 annual report disclosed that Azure’s annual revenue exceeded \$75 billion for the first time.

AWS has long been a core profit engine for Amazon. In Q1 2026, AWS operating income reached \$14.2 billion.

Nvidia is an even clearer example. In Q1 fiscal year 2026, Nvidia’s Data Center revenue reached \$75.2 billion, up 92% year over year. This shows that AI data centers and compute infrastructure have become one of the most important battlefields in the current technology industry.

Therefore, the core departments of major tech companies today must include:

Cloud Platform, AI Infrastructure, GPU Infrastructure, Distributed Systems, Storage, Networking, Serving Infrastructure, Database, Observability, and Developer Platform.

## **6. AI / ML Platform Has Evolved from a “Research Department” into a Core Production Department**

In the past, AI Research may have been closer to frontier research. Today, it has become a core production department that directly serves business needs.

Core areas include:

LLMs, Recommendation Models, Ads Models, Search Models, Foundation Models, AI Agents, AI Coding, AI Safety, Model Serving, Training Infrastructure, Inference Optimization, Evaluation, RAG, and Multimodal AI.

Google, Microsoft, Meta, Amazon, and Apple are all investing heavily in AI infrastructure. For example, Alphabet has repeatedly emphasized AI infrastructure investment and growing demand for AI services, while the market is also paying close attention to its AI capital expenditures.

So from a job-seeking or training perspective, the areas most worth emphasizing today are:

AI Infrastructure + Applied ML + Product AI + AI Platform.

These are much closer to the real core businesses of major tech companies than pure model research alone.

## **7. Data / Experimentation / Analytics Is the Decision-Making System of Major Tech Companies**

These departments may not directly generate the most revenue, but they serve as the “decision-making center” of major tech companies.

Core teams include:

Data Science, Product Analytics, Experimentation Platform, A/B Testing Platform, Causal Inference, Metrics Platform, Data Engineering, Data Warehouse, Business Intelligence, Forecasting, and Measurement.

Data Scientist, Data Engineer, and Analytics Engineer roles are especially important in departments such as Ads, Growth, Product, Recommendation, Marketplace, and Subscription.

In real major tech companies, many product decisions are not based on intuition. They rely on:

Metric design, A/B testing, incrementality, causal analysis, cohort analysis, retention analysis, and revenue attribution.

Therefore, from a role perspective, Data Science is most valuable in Ads, Growth, Product, Marketplace, Subscription, and Recommendation.

## **8. Trust / Safety / Privacy / Security Is an Increasingly Core “Moat Department”**

In the past, these departments may have been seen as support functions. That is no longer the case.

The reason is simple:

Advertising privacy, content safety, AI safety, data compliance, child safety, anti-abuse, account security, payment risk control, and platform governance all directly affect whether a company can continue operating.

Core areas include:

Privacy Engineering, Trust & Safety, Integrity, Anti-abuse, Fraud Detection, Content Moderation, Security Engineering, Risk Platform, Compliance Engineering, AI Safety, and Guardrails.

Meta, Google, TikTok, Apple, and Microsoft all place significant importance on these areas. The stronger advertising and recommendation systems become, the more regulatory pressure they face, and the more core these departments become.

## **9. Marketplace / Commerce / Payment / Logistics Are Core Business Lines for Amazon, Apple, Google, and TikTok**

Not every company has these business lines, but they are especially important for Amazon and Apple.

Amazon's core is not just its e-commerce webpage. Its core includes:

Marketplace, Seller Platform, Search Ranking, Recommendation, Pricing, Fulfillment, Logistics, Prime, Ads, Payments, and Fraud.

Apple's core is not just iPhone hardware. It also includes:

iOS, App Store, Apple Pay, Services, Subscriptions, iCloud, Apple Music, and Apple TV+.

Apple disclosed in Q4 2025 that Services revenue reached a record high, showing that Services has become a very important high-margin business outside of hardware.

## **10. Developer Ecosystem / Platform Is a Core Moat for Microsoft, Apple, Google, and Nvidia**

The value of these departments is not short-term revenue, but locking in the developer ecosystem.

Typical examples include:

Microsoft: Azure, GitHub, VS Code, Windows, Copilot, .NET

Apple: iOS SDK, App Store, Xcode, Apple Silicon

Google: Android, Chrome, Google Cloud, TensorFlow/JAX, Vertex AI

Nvidia: CUDA, GPU software stack, AI Enterprise, TensorRT

Nvidia is the most typical example. It is not just selling GPUs. It has built a massive moat through CUDA, AI libraries, and its developer ecosystem. The rapid growth of its Data Center revenue also shows that the AI compute ecosystem has become a core business.

## **The Most Realistic Ranking of Core Departments in Major Tech Companies**

If ranked by general importance, I would rank them as follows:

### **Tier 1: Company Lifeline Departments**

1. Ads / Monetization
2. Search / Feed / Recommendation / Ranking
3. Core Product
4. Cloud / Infrastructure
5. AI / ML Platform

These departments are the closest to revenue, users, technical barriers, and company strategy.

### **Tier 2: Strong Core Departments**

1. Growth
2. Data Science / Experimentation
3. Marketplace / Commerce / Payment
4. Trust / Safety / Privacy / Security
5. Developer Platform / Ecosystem

These departments may not be the absolute top priority at every company, but they are generally very important inside major tech companies.

### **Tier 3: Core Support Departments**

1. Enterprise Sales
2. Customer Success
3. Legal / Policy
4. Finance / Strategy
5. HR / Recruiting
6. Operations

These departments are also important, but they are usually not the “technical core departments” of major tech companies.

# A More Realistic Breakdown by Company

## Google / Alphabet

Most core departments:

Search, Ads, YouTube, Google Cloud, Android, Chrome, Maps, AI/DeepMind, and Core Infrastructure.

Google's cash flow core is still Search + Ads, while its strategic core is AI + Cloud + YouTube.

## Meta

Most core departments:

Facebook/Instagram Feed, Reels, Ads, Recommendation, Family of Apps, WhatsApp, AI Infrastructure, and Reality Labs.

However, from a revenue perspective, Meta's most core business is still Family of Apps advertising. Reality Labs is a strategic exploration area, not the current profit core.

## Amazon

Most core departments:

AWS, Marketplace, Retail Search, Ads, Prime, Logistics, Fulfillment, Pricing, and Recommendation.

Amazon is one of the few companies that simultaneously owns four major core systems: e-commerce, cloud, advertising, and logistics.

## Microsoft

Most core departments:

Azure, Microsoft 365, Copilot, Windows, GitHub, Developer Tools, Enterprise Cloud, and Security.

Microsoft's core is not consumer apps, but enterprise software, cloud, AI, and the developer ecosystem.

## Apple

Most core departments:

iPhone, iOS, Apple Silicon, App Store, Services, Wearables, Privacy, and Ecosystem.

Apple's core is a closed loop of hardware + operating system + services ecosystem.

## **Netflix**

Most core departments:

Content, Recommendation, Streaming Infrastructure, Membership, Pricing, Ads, and Personalization.

Netflix's recommendation system is very important, but content and the subscription business model are equally core.

## **TikTok / ByteDance**

Most core departments:

Recommendation, Feed Ranking, Ads, Creator Ecosystem, E-commerce, Content Safety, Growth, and AI Infrastructure.

TikTok is the most typical example of an algorithm-driven product.